NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

THEORY EXAMINATION

December 2025

Roll No._____

Programme	:	B. Tech.	Semester	:	3rd
Subject	:	Mechanics of Machines	Max. Time	:	3 hrs.
Subject code	:	RAPC 201	Max. Marks.	:	50
Total pages	:	02			

Important Instructions:

a) The candidates, before starting to write the answers, should please check the Question Paper for any discrepancy and also ensure that they have been delivered the question paper of right course number and right subject title.

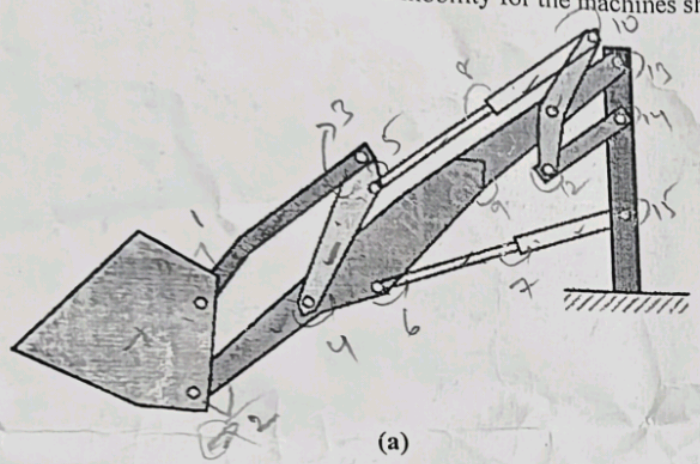
b) **Attempt all questions.**

c) Clearly state assumptions wherever taken. Assume suitable data wherever required.

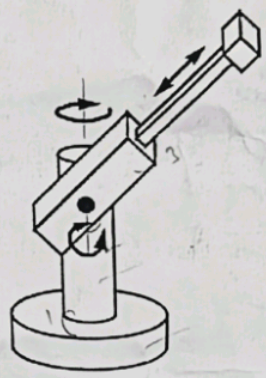
Q1. (a) Define kinematic link, kinematic pair, kinematic chain, machine and mechanism [5]

(b) What do you mean by kinematic inversion? Discuss various inversions of a slider-crank mechanism. [5]

Q2. (a) Determine the number of links and mobility for the machines shown below (**Fig. Q2**) [5]



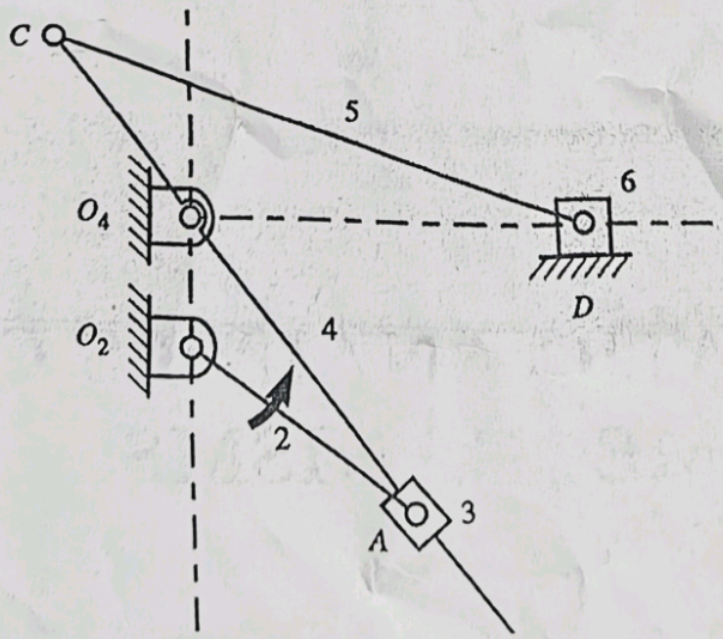
(a)



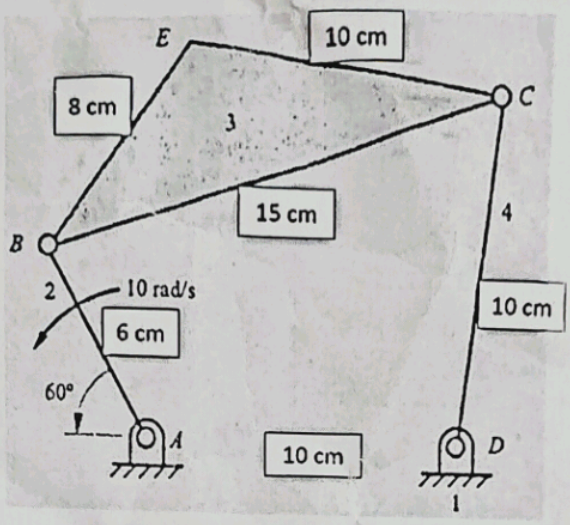
(b)

Fig. Q2

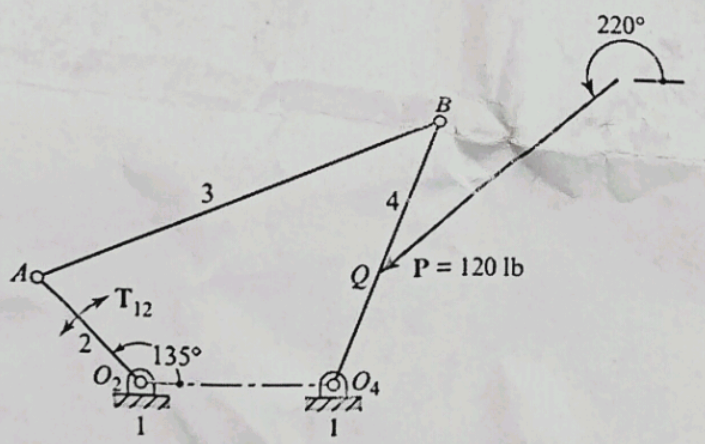
(b) Figure below shows a six-link, Whitworth quick-return mechanism used in slotting machines. Link 2 (= O_2A) rotates with a constant angular speed in the counter-clockwise (CCW) direction. The sliding link 6 represents the cutting tool whose cutting motion is to the left and the return (idle) motion is to the right. Determine the quick-return ratio which is defined as the ratio of the time intervals taken by the tool to complete its cutting and idle motions, respectively. [5]

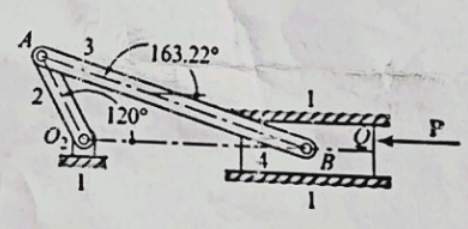


Q3. Determine the angular positions, angular velocities, and angular accelerations of all members of the linkage shown below when link AB is at 60° to the horizontal. Using graphical analysis, also find the position, velocity, and acceleration of point E in the coupler member of the linkage. Link AB is driven at a constant angular velocity of 10 rad/s counter clock-wise. [10]



Q4. (a) For the mechanisms shown below, draw the free-body diagram of moving links. [5]





(b) Discuss the difference between open-belt drive and crossed-belt drive. [5]

Q5. (a) Two parallel shafts, connected by a crossed belt, are provided with pulleys 480 mm and 640 mm in diameters. The distance between the centre lines of the shafts is 3 m. Find by how much the length of the belt should be changed if it is desired to alter the direction of rotation of the driven shaft. [5]

(b) What is the effect of centrifugal tension on the power transmitted in a belt drive [5]